

Fig. 5.39

Example 5.27

The orbit of Halley's Comet (Fig. 5.51) is an ellipse 36.18 astronomical units long and by 9.12 astronomical units wide. Find its eccentricity.

Solution

Given that $2a = 36.18,\ 2b = 9.12$, we get

$$e = \sqrt{1-\frac{b^2}{a^2}} = \frac{\sqrt{a^2-b^2}}{a} = \frac{\sqrt{\left(\frac{36.18}{2}\right)^2 - \left(\frac{9.12}{2}\right)^2}}{\frac{36.18}{2}}$$

$$= \frac{\sqrt{(18.09)^2-(4.56)^2}}{(8.09)} \approx 0.97 .$$

Note

One astronomical unit (mean distance of Sun and earth) is $1,49,597,870\,km$, the semi major axis of the Earth's orbit.

EXERCISE 5.2

1. Find the equation of the parabola in each of the cases given below:
   (i) focus $(4,0)$ and directrix $x = -4$.
   (ii) passes through $(2,-3)$ and symmetric about y-axis.
   (iii) vertex $(1,-2)$ and focus $(4,-2)$.
   (iv) end points of latus rectum $(4,-8)$ and $(4,8)$.
2. Find the equation of the ellipse in each of the cases given below:
   (i) foci $(\pm 3,0), e = \frac{1}{2}$.
   (ii) foci $(0,\pm 4)$ and end points of major axis are $(0,\pm 5)$.
   (iii) length of latus rectum 8, eccentricity $= \frac{3}{5}$, centre (0, 0) and major axis on x-axis.
   (iv) length of latus rectum 4, distance between foci $4\sqrt{2}$, centre (0, 0) and major axis as y-axis.
3. Find the equation of the hyperbola in each of the cases given below:
   (i) foci $(\pm 2,0)$, eccentricity $= \frac{3}{2}$.
   (ii) Centre $(2,1)$, one of the foci $(8,1)$ and corresponding directrix $x = 4$.
   (iii) passing through $(5,-2)$ and length of the transverse axis along x axis and of length 8 units.

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